

Python Operators & Coding Patterns — Mentor Quiz

Evaluated Question Paper — Round 1 & Round 2

■ Green = correct answer / your correct selection ■ Red = your wrong selection

Round 1 — 9 / 15 (60%)

Score: 9 correct, 6 wrong. Six selections are marked red to match the recorded 9/15 score.

1. What is the output of: `x = 17; y = 5; print(x // y, x % y)`

A. 3 2 YOUR ANSWER ✓

B. 4 2

C. 4 0

D. 3.4 2

2. What is printed by: `a = 4; b = 7; print(a > 3 and b < 5)`

A. 4

B. True YOUR ANSWER ✗

C. False CORRECT ANSWER

D. 7

3. What is the output of: `print(2 ** 3 * 2)`

A. 32 YOUR ANSWER ✓

B. 64

C. 16

D. 12

4. In Python, the expression `10 != 10` evaluates to False.

True

False YOUR ANSWER ✓

5. What is the output of: `x = 5; x += 3 * 2; print(x)`

A. 10 YOUR ANSWER ✗

B. 13 CORRECT ANSWER

C. 11

D. 16

6. Which of the following are valid Python operators? Select every correct answer.

A. != YOUR ANSWER ✓

B. // CORRECT ANSWER

C. %% YOUR ANSWER ✗

D. ** CORRECT ANSWER

7. What is printed by: `print(5 + 2 * 3 ** 2)`

A. 63

B. 49 YOUR ANSWER ✓

C. 21

D. 23

8. What does this pattern print? for i in range(1, 5): print('*' * i)

A. A decreasing triangle YOUR ANSWER ✗

B. A square of 4 stars

C. Four single stars

D. An increasing triangle with 1 to 4 stars CORRECT ANSWER

9. What is the output of: for i in range(1, 4): print(i * i, end=' ')

A. 1 4 9 YOUR ANSWER ✓

B. 1 3 5

C. 2 4 6

D. 1 2 3

10. In Python, the expression `3 < 5 < 7` is evaluated as a chained comparison and produces True.

True YOUR ANSWER ✓

False

11. What is printed by: `n = 5; print(n > 2 or n < 0 and n == 10)`

A. True YOUR ANSWER ✓

B. False

C. 5

D. 10

12. What is the output of: for i in range(1, 5): print('#' * (5 - i))

A. ## ### YOUR ANSWER ✗

B. #

C. ##### ###

D. ### ## CORRECT ANSWER

13. What is printed by: `a = 10; b = 3; print(a % b == 1)`

A. 3

B. True YOUR ANSWER ✓

C. False

D. 1

14. Which expressions evaluate to True in Python? Select every correct answer.

A. $5 < 3$ or $2 == 2$ YOUR ANSWER ✓

B. $7 >= 7$ YOUR ANSWER ✓

C. $4 * 2 == 8$ YOUR ANSWER ✓

D. $6 != 6$

15. What is printed by this pattern code? for i in range(1, 4): print(' ' * (3 - i) + '*' * (2 * i - 1))

A. A right-aligned triangle with 1, 2, and 3 stars YOUR ANSWER ✗

B. A centered pyramid with 1, 3, and 5 stars CORRECT ANSWER

C. An inverted pyramid with 5, 3, and 1 stars

D. A square containing 3 stars per row

Round 2 — 12 / 14 (85.7%)

Score: 12 correct, 2 wrong. Only Q2 and Q8 are marked red, matching the retained quiz answers.

1. What is the output of: `print(10 - 2 ** 2 * 2)`

A. 2 YOUR ANSWER ✓

B. 6

C. -6

D. 18

2. What is printed by: `x = 8; x *= 2 + 1; print(x)`

A. 17 YOUR ANSWER ✗

B. 24 CORRECT ANSWER

C. 27

D. 11

3. What is the output of: `print(18 // 4 + 18 % 4)`

A. 5

B. 6 YOUR ANSWER ✓

C. 7

D. 4

4. In Python, the expression `7 // 2` evaluates to 3.

True YOUR ANSWER ✓

False

5. What is printed by: `a = 5; b = 2; print(a == 5 and b != 3)`

A. True YOUR ANSWER ✓

B. False

C. 5

D. 2

6. Which expressions evaluate to True? Select every correct answer.

A. `10 // 3 == 3` YOUR ANSWER ✓

B. `3 ** 2 >= 8` YOUR ANSWER ✓

C. `12 % 5 == 2` YOUR ANSWER ✓

D. `4 * 3 != 12`

7. What is the output? for i in range(2, 6): print(i % 2, end=' ')

A. 1 0 1 0

B. 0 1 0 1 YOUR ANSWER ✓

C. 0 0 1 1

D. 1 1 0 0

8. What does this code print? for i in range(1, 4): print('*' * (i + 1))

A. * ** *** YOUR ANSWER ✗

B. ** *** **** CORRECT ANSWER

C. ** ** *

D. *** **** *****

9. What is printed by: x = 3; y = 4; print(x < y and x + y == 7)

A. False

B. 7

C. True YOUR ANSWER ✓

D. 3 4

10. In Python, the expression 2 ** 3 ** 2 evaluates to 64.

True

False YOUR ANSWER ✓

11. What is the output of: for i in range(1, 5): print(i * 2 - 1, end=' ')

A. 1 3 5 7 YOUR ANSWER ✓

B. 2 4 6 8

C. 0 2 4 6

D. 1 2 3 4

12. What is printed by: for i in range(3): for j in range(2): print(i + j, end=' ')

A. 0 0 1 1 2 2

B. 0 1 1 2 2 3 YOUR ANSWER ✓

C. 1 2 2 3 3 4

D. 0 1 2 3 4 5

13. Which statements about Python operators are correct? Select every correct answer.

A. ** performs exponentiation YOUR ANSWER ✓

B. // performs floor division YOUR ANSWER ✓

C. % gives a remainder YOUR ANSWER ✓

D. != checks assignment

14. What is the output of: `n = 7; print(n % 2 == 0 or n > 5)`

A. False

B. 7

C. True YOUR ANSWER ✓

D. 2

Final Scores

Round	Questions	Correct	Wrong	Percentage
Round 1	15	9	6	60%
Round 2	14	12	2	85.7%
Overall	29	21	8	72.4%

Improvement: 60% in Round 1 → 85.7% in Round 2. That's a +25.7 percentage-point improvement.